Name (Last, First):

BU ID:

Assignment 2

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 108 = 3 + 105 pts]**

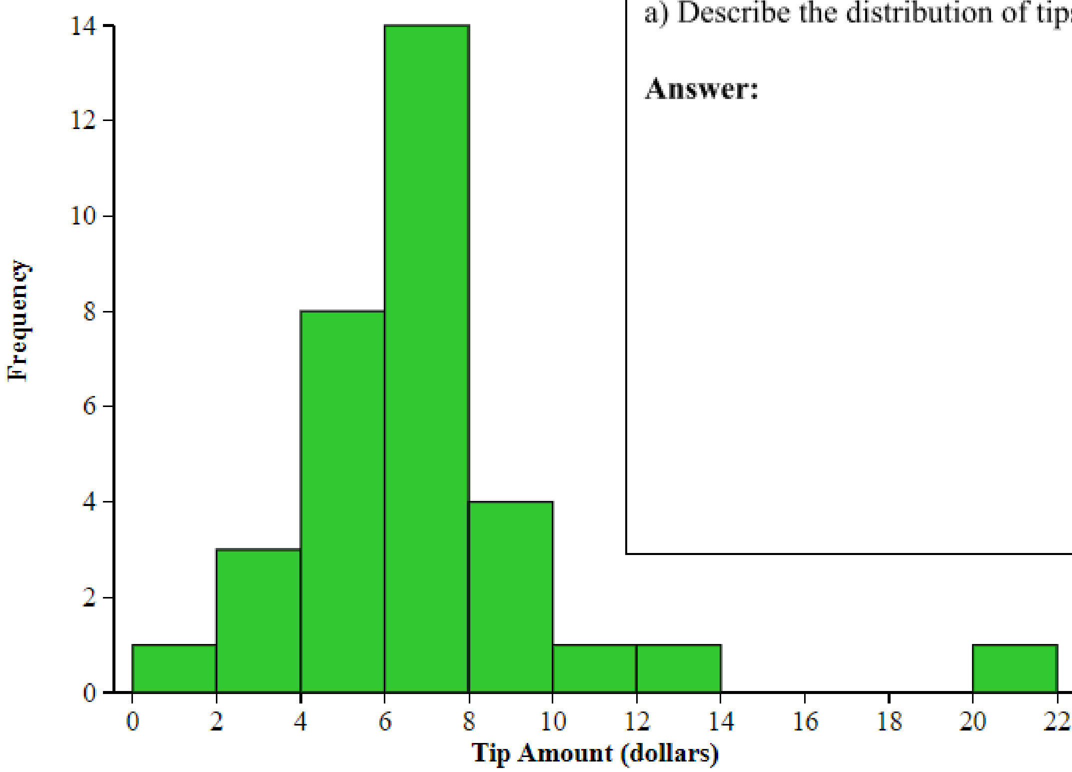
Grading Rubric

Each question is worth 3 points and will be graded as follows:
3 points: Correct answer with work shown
2 points: Incorrect answer but attempt shows some understanding (work shown)
1 point: Incorrect answer but an attempt was made (work shown)
0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]
(link to your live google doc)

I. SOCS and Comparing (33 pts = 11 x 3 pts)

1) Karley works as a Dominoes delivery driver and on Friday night, she recorded how many tips she got from each of her deliveries. The histogram below shows the distribution of the 33 deliveries.



a) Describe the distribution of tips.

Answer:

b) One of the tip amounts was $9. If this tip had been $14 instead of $9, describe what effect this would have on the following measures of the center. Justify your answer.

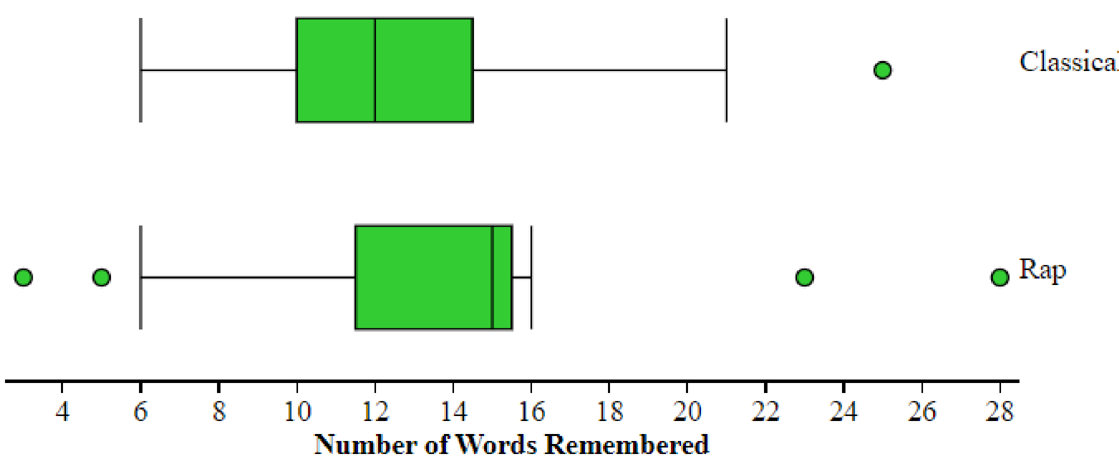
Mean:

Median:

c) Describe how you would estimate the median of this distribution.

Answer:

2) Jordan and Alex want to know if listening to different types of music while studying will help you remember the material better. They randomly assigned a group of students to two different groups: one group studied a list of words while listening to classical music while the second group studied the same list of words while listening to rap music. There was a total of 30 words on the list and each group studied the list while listening to the music for 5 minutes. They were then asked to write down as many words as they could remember. The data is displayed in the boxplots below.



a) Approximate the interquartile range for each set of data. Why is this the appropriate measure of spread to use for these two data sets?

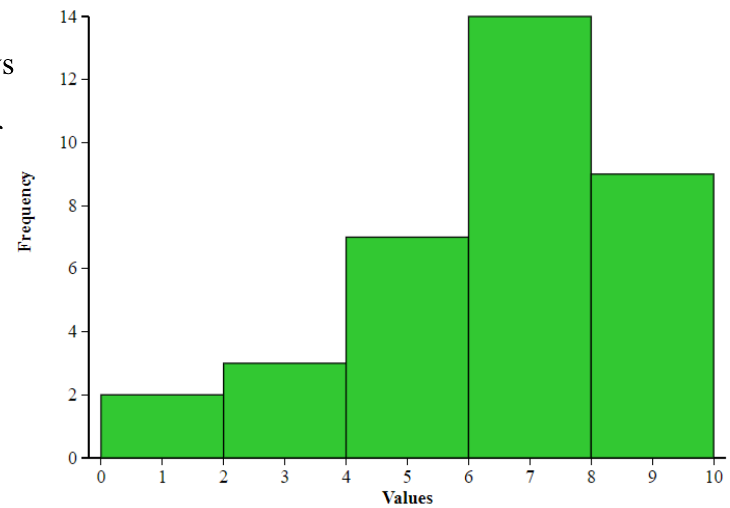
Answer:

b) Write three sentences comparing the number of words remembered between each of the two groups.

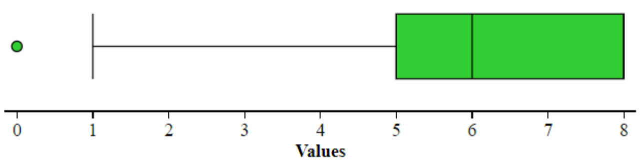
Answer:

Multiple Choice Practice

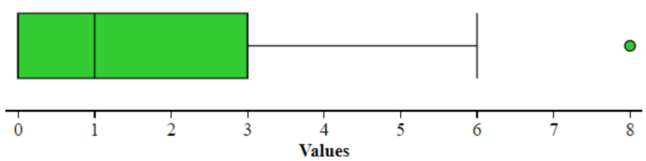
3) The histogram below displays a set of measurements. Which of the boxplots below displays the same set of measurements?



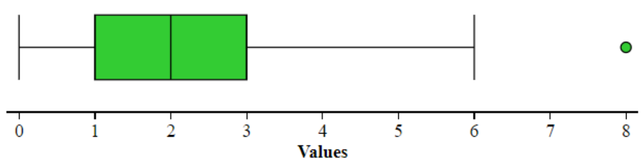
(A)



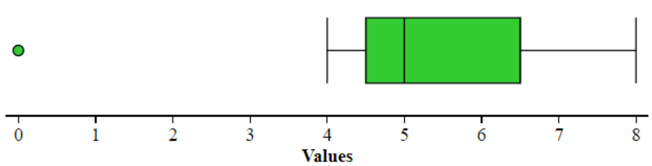
(B)



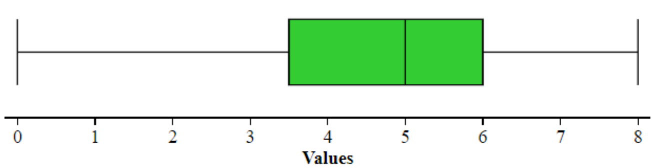
(C)



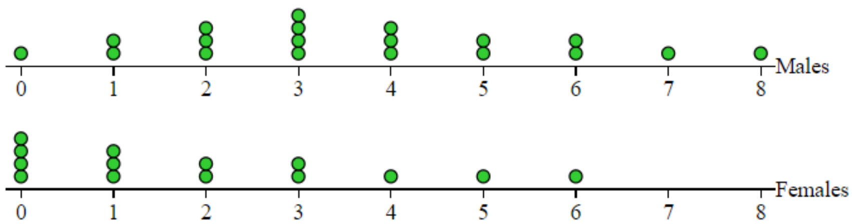
(D)



(E)



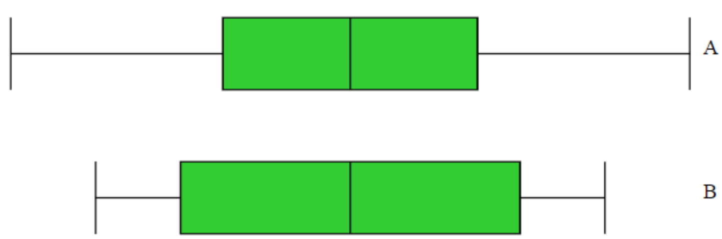
______4) These dotplots for randomly selected male and female students at a particular high school show the number of times per week they eat at fast food restaurants.



Which of the following is a true statement?

(A) One distribution is roughly symmetric; the other is skewed left.
(B) The males' median is larger than the females' mean.
(C) The ranges are both 8.
(D) The standard deviations are equal.
(E) Combining the male and female times into one set of student times will increase the range to 16.

______5) The boxplots below summarize two sets of data, A and B. Which of the following must be true?



I. Set A contains more data than Set B.
II. Set A has a larger range than Set B.
III. Set A and Set B have the same median.

(A) I only
(B) III only
(C) I and II only
(D) II and III only
(E) I, II, and III

_ 6) Which of the following statements is incorrect?

(A) Both dotplots and stemplots can show symmetry, gaps, clusters, and outliers.
(B) Sets with different distribution shapes can have identical boxplots.
(C) Boxplots, dotplots, stemplots, and histograms can all show skewness.
(D) In histograms, relative areas correspond to relative frequencies.
(E) In histograms, frequencies can be determined from relative heights.

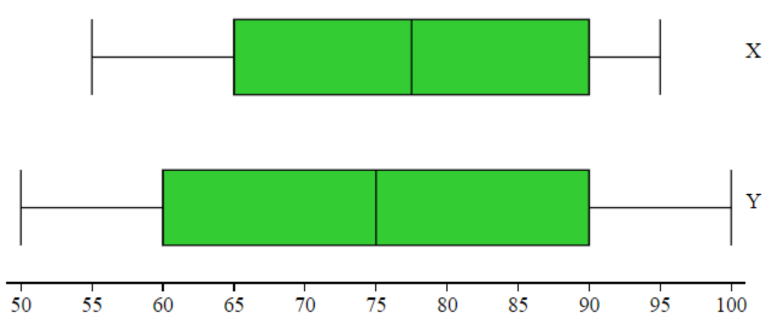
_ 7) Consider the following test scores:

Class A: 65, 72, 73, 73, 74, 82, 91 Class B: 60, 65, 79, 81, 85, 86, 86

Which of the following is a true statement?

(A) Class A has the greater range, while Class B has the greater standard deviation.
(B) Class B has the greater range, while Class A has the greater standard deviation.
(C) The ranges and standard deviations are the same for both classes.
(D) The ranges are the same, but Class A has the higher standard deviation.
(E) The ranges are the same, but Class B has the higher standard deviation.

8) The boxplots shown below summarize two sets of data, X and Y.



Which of the following must be true?

(A) 50 is an outlier for Set Y.
(B) Set Y contains more data than Set X.
(C) The IQR of Set X is smaller than the IQR of Set Y.
(D) The range of Set Y is smaller than the range of Set X.
(E) The means of the two data sets are equal.

II. The Empirical Rule and Z-Scores (72 pts = 24 x 3 pts)

1) Larry came home very excited after a visit to his doctor. He announced proudly to his wife, "My doctor says my blood pressure is at the 90th percentile among men like me. That means I'm better off than about 90% of similar men." How should his wife, who is a statistician, respond to Larry's statement?

Answer:

2) The distribution of heights of adult American men is approximately normal with mean 69 inches and standard deviation 2.5 inches. Draw a normal curve below and mark three standard deviations out on either side. Then, use the curve to answer the following questions.

Answer:

(a) What percent of men are shorter than 64 inches?

(b) What percent of men are taller than 71.5 inches?

(c) What percent of men are taller than 66.5 inches?

(d) A height of 74 inches corresponds to what percentile of American male heights?

3) The length of human pregnancies from conception to birth varies on an approximate normal distribution with mean 266 days and a standard deviation of 6 days. Draw a normal curve below and mark three standard deviations out on either side. Then, use the curve to answer the following questions.

a) How long are the longest 16% of pregnancies?

b) How short are the shortest 16% of pregnancies?

c) Between how many days does the middle 95% of all pregnancies fall?

d) Between how many days does the middle 99.7% of all pregnancies fall?

e) A pregnancy of 254 days corresponds to what percentile of pregnancies?

4) Three landmarks of baseball achievement are Ty Cobb's batting average of 0.420 in 1911, Ted Williams's 0.406 in 1941, and George Brett's 0.390 in 1980. These batting averages cannot be compared directly because the distribution of major league batting averages has changed over the years. The distributions are quite symmetric, except for outliers such as Cobb, Williams, and Brett. While the mean batting average has been held roughly constant by rule changes and the balance between hitting and pitching, the standard deviation has dropped over time. Here are the facts:

Decade	Mean	Standard Deviation
1910s	0.266	0.0371
1940s	0.267	0.0326
1970s	0.261	0.0317

Find the standardized scores for Cobb, Williams, and Brett. Who was the best hitter?

5) The scores of an introductory statistics class at the University of Wisconsin is normally distributed with a mean of 72 and standard deviation of 10.

(a) If a student has a z-score of 1.4, what score does she have?

(b) If a student has a z-score of -1.23, what score does he have?

6) A college statistics exam scores were normally distributed with $\mu = 150$ points and $\sigma = 20$ points. The z-scores for some students are below. Find the final exam raw score for each.

Chris 0.1

Janet 1.8

Rick -1.1

7) Make an accurate sketch of a Standard Normal Distribution and answer each question using a standard normal curve.

a) What is the area between $z = 0$ and $z = 1$?

b) What percent of the data is within one standard deviation of the mean?

c) What percent of the data is more than one standard deviation from the mean?

d) What is the area between $z = 0$ and $z = 2$?

e) What is the area under the curve between $z = -1$ and $z = 2$?

THE END